Medical Robotics

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Lecture 10
Robot Registration

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Contents

1. Lecture Overview

This lecture provides a complete treatment of **robot registration** — the process of determining the geometric relationships between reference frames that allow a pre-operative surgical plan to be transferred to the coordinate system of the robot.

Following a review of definitions and the three operation types (calibration, tracking, and registration), the mathematical formulation of rigid registration is recalled and the two main 3D/3D registration paradigms — **point-to-point** and **point-to-surface** — are derived. The **Iterative Closest Point (ICP)** algorithm is described in detail as the standard solver for point-to-surface registration. Six concrete clinical examples are then analysed (Robodoc, CASPAR, Acrobot, Speedy, CyberKnife, CyberKnife+Synchrony), illustrating how the general framework specialises to different surgical domains and hardware configurations. The lecture concludes with a description of a current research project on **physical–digital twins** for surgical training and planning, in which the registration methods covered in this lecture are applied to match a physical anatomical phantom with its digital counterpart in real time.

> **Connection to Lecture 9.** The need for registration was introduced at the end of Lecture 9 as an enabling technology for virtual fixture-based surgical assistance. This lecture provides the full technical treatment. The key message remains the same: a VF or autonomous robot trajectory defined in the pre-operative imaging frame must be expressed in the robot's frame before it can be executed, and this transformation is what registration computes.

2. Definitions and Operation Types

> **Registration.** The process of determining the geometric relationship (i.e., the rigid transformation: rotation and translation) between two or more coordinate reference frames.
>
> **Robot registration.** The specific instance of registration whose purpose is to determine the geometric relationships between reference frames that allow the transfer of a digital surgical plan to the coordinate system of the robot.

The operations involved in building and maintaining the network of reference frames in an operating room are of three distinct types, which must not be confused:

Calibration (C). Establishes the relationship between the reference frames of devices that are co-located in the operating room in a stable, long-term configuration (e.g., robot base frame ↔ optical localizer frame, C-arm ↔ localizer frame). Calibration is performed once, or whenever the room setup changes, and the result is stored. It is the same type of procedure as the intrinsic kinematic calibration of the robot itself, which establishes the precise link and joint parameters of the robot's kinematic chain.

Tracking (T). Continuously updates the transformation between a moving tool's reference frame and the localizer frame, in real time, at the speed of the localizer's measurement rate. The localizer observes the positions of markers mounted on the tool and reconstructs the tool's pose at each time step.

Registration (R). Determines the transformation between the pre-operative imaging frame (in which the surgical plan is expressed) and the intra-operative reference frame (in which the robot and localizer operate). Registration is performed ideally once at the start of each procedure, but may need to be repeated if the patient moves.

> **Multi-modal image registration.** In some procedures, the pre-operative plan itself is the result of a registration procedure: CT images (best for bone) and MRI images (best for soft tissue) may be fused to build a more complete anatomical model. Registering images from different modalities, or from the same modality at different times (e.g., to track disease evolution over months or years), is a separate and often more complex problem because the image appearance of the same anatomy differs across modalities and times. Rigid registration suffices for bony anatomy; soft tissue and growing patients require non-rigid (deformable) registration methods.

3. Devices and Reference Frames

3.1. Localizers

Localizers determine the pose of surgical instruments and anatomical references in the operating room. Three generations of localizer technology exist:

Optical localizers (current standard of care). An infrared camera system tracks retroreflective passive markers (or active infrared-emitting diodes) mounted in a specific geometric configuration on the tool. By detecting the 2D positions of the markers in the camera images and knowing their 3D geometry, the pose of the tool rigid body is reconstructed. The NDI Polaris is the most widely used clinical system. Its main limitations are a restricted field of view and susceptibility to occlusion: if the line of sight between the camera and any marker is blocked, tracking is lost.

Magnetic localizers. A magnetic field generator produces a spatially varying field; sensors embedded in the tool measure the local field strength and direction to compute the tool's pose. Advantages: no line-of-sight requirement, low cost, small sensor size. Disadvantage: sensitivity to electromagnetic interference from other OR equipment (surgical energy devices, motors), which limits clinical adoption. Currently a research topic.

Mechanical localizers. Passive articulated arms with joint encoders, mounted to a fixed point in the OR. The joint angles are read to compute the pose of the tool attached at the tip via forward kinematics. These were used in early systems (e.g., the ISG Viewing Wand in neurosurgery) but are now largely abandoned because they are cumbersome and restrict the surgeon's movements.

3.2. Intra-operative Imaging Sensors

Real-time intra-operative imaging is used to acquire data for registration and to verify tool placement during the procedure. The main modalities are fluoroscopy (C-arm X-ray), digital radiography, and ultrasound. Fluoroscopy provides real-time 2D projective images of bony anatomy and is the most widely used. Ultrasound is radiation-free and suitable for soft tissue, but image quality is operator-dependent and resolution is lower.

3.3. Reference Frames of Interest

Frame	Description
$\mathcal{R}_{\text{patient/planning}}$	Patient anatomy as represented in the pre-operative imaging system (CT, MRI). The surgical plan is defined here.
$\mathcal{R}_{\text{localizer/OR}}$	Fixed frame of the optical localizer in the operating room.
$\mathcal{R}_{\text{patient/OR}}$	Patient anatomy as it lies on the OR table during surgery. Differs from $\mathcal{R}_{\text{patient/planning}}$ due to patient repositioning.
$\mathcal{R}_{\text{img\_sensor/OR}}$	Frame of the intra-operative imaging device (C-arm, ultrasound probe).
$\mathcal{R}_{\text{robot/OR}}$	Base frame of the surgical robot in the OR.

The chain of transformations linking these frames is what robot registration must establish. Note that all intra-operative frames exist simultaneously during surgery, while $\mathcal{R}_{\text{patient/planning}}$ belongs to the pre-operative phase and must be related to the intra-operative frames via registration.

4. Mathematical Formulation of Rigid Registration

4.1. General Problem Statement

Given two reference frames $\mathcal{R}_A$ and $\mathcal{R}_B$, rigid registration seeks the homogeneous transformation matrix ${}^AT_B$ (comprising rotation ${}^AR_B \in SO(3)$ and translation ${}^A\boldsymbol{t}_B \in \mathbb{R}^3$) such that the coordinates of the same physical feature expressed in the two frames are related by:

$$
{}^A\boldsymbol{f} = {}^AR_B\,{}^B\boldsymbol{f} + {}^A\boldsymbol{t}_B
$$

The procedure is:

1. **Select features.** Identify a set of n corresponding features $\{{}^A\boldsymbol{f}_i\}_{i=1}^n$ in frame $\mathcal{R}_A$ and $\{{}^B\boldsymbol{f}_i\}_{i=1}^n$ in frame $\mathcal{R}_B$.
2. **Define a distance measure.** Choose a similarity or distance metric $\mathcal{D}$ between the features in $\mathcal{R}_A$ and the transformed features from $\mathcal{R}_B$.
3. **Optimise.** Find the transformation that minimises the distance:

$$
{}^AT_B^* = \arg\min_{{}^AT_B} \mathcal{D}\big({}^A\boldsymbol{f},\ {}^AT_B\,{}^B\boldsymbol{f}\big)
$$

For point features with the standard sum-of-squared-distances metric, this becomes:

$$
({}^AR_B^*,\ {}^A\boldsymbol{t}_B^*) = \arg\min_{R,\boldsymbol{t}} \sum_{i=1}^{n} \left\| {}^A\boldsymbol{f}_i - R\,{}^B\boldsymbol{f}_i - \boldsymbol{t} \right\|^2
$$

This is a least-squares problem over $SO(3) \times \mathbb{R}^3$. The minimum number of non-collinear point pairs required to determine a unique rigid transformation is $n = 3$; in practice $n \gg 3$ is used to reduce sensitivity to measurement noise.

The features $\boldsymbol{f}_i$ can be:

- **Artificial (fiducial markers)**: radio-opaque or optically visible markers screwed into bone or attached to tissue. They provide reliable, easily localised feature points but require an invasive placement step.
- **Anatomical landmarks**: naturally identifiable points in both the pre-operative image and the intra-operative scene (e.g., bone prominences, vessel bifurcations). Non-invasive, but often less precise.

Minimum number of points. Three non-collinear corresponding point pairs uniquely determine the 6-DOF rigid transformation (3 for rotation, 3 for translation). Using more points over-determines the system and the least-squares solution averages out measurement noise. Points must not be collinear (all on a single line) or coplanar in degenerate configurations; in practice, fiducials are distributed in 3D to maximise geometric diversity.

4.2. 3D/3D Registration Methods

The two principal categories of 3D/3D registration used in surgical robotics are:

Point-to-point registration (Procrustes problem). A one-to-one correspondence is established between $n \geq 3$ points in frame $\mathcal{R}_A$ and the same n points in frame $\mathcal{R}_B$. Correspondences are known *a priori* (each point in one frame is labelled and matched to its counterpart in the other). The optimal rigid transformation can then be computed in closed form using singular value decomposition (SVD). Correspondences are established via external fiducial markers (screwed into bone or attached to skin) or anatomical landmarks identified manually in both frames.

Point-to-surface registration. A set of points is acquired intra-operatively (e.g., by palpating the bone surface with a tracked pointer) and must be matched to a 3D surface model reconstructed from pre-operative images. Correspondences are *not* known in advance: each intra-operative point must be matched to its closest point on the pre-operative surface. The standard algorithm for this problem is the **Iterative Closest Point (ICP)** method. Point-to-surface registration is non-invasive (no fiducials must be implanted) but requires a good initial estimate and is more sensitive to local minima than point-to-point registration.

5. The Iterative Closest Point (ICP) Algorithm

5.1. Problem Statement

Given:

- A **data shape** P: a set of N_p points acquired intra-operatively (e.g., by palpating the exposed bone surface with a tracked stylus).
- A **model shape** M: a surface (set of N_m supporting points, triangles, or other geometric primitives) reconstructed from pre-operative CT images.

Find the rigid transformation T that, when applied to P, provides the best alignment of P with M.

The key insight of ICP is: *under certain conditions, the point correspondences provided by nearest-neighbour matching (closest-point pairs) are a reasonable approximation to the true correspondences.* This allows the alternation of two steps — correspondence estimation and transformation estimation — until convergence.

5.2. Algorithm

Starting from an initial estimate ${}^{A}T_{B}^{(0)}$ (which must be reasonably close to the true transformation to avoid convergence to a wrong local minimum):

1. **Transform.** Express the intra-operative points ${}^{B}\boldsymbol{p}_i$ in the model frame using the current estimate:
$$ {}^{A}\tilde{\boldsymbol{p}}_i^{(k)} = {}^{A}T_{B}^{(k)}\,{}^{B}\boldsymbol{p}_i $$

2. **Find closest points.** For each transformed point ${}^{A}\tilde{\boldsymbol{p}}_i^{(k)}$, find its closest point on the model surface M:
$$ {}^{A}\boldsymbol{m}_i^{(k)} = \text{closest}\Big({}^{A}\tilde{\boldsymbol{p}}_i^{(k)},\ M\Big) $$

3. **Update transformation.** Solve the point-to-point registration problem with the current correspondences $\{({}^{A}\boldsymbol{m}_i^{(k)},\ {}^{B}\boldsymbol{p}_i)\}$:
$$ {}^{A}T_{B}^{(k+1)} = \arg\min_{T} \sum_{i=1}^{N_p} \left\| {}^{A}\boldsymbol{m}_i^{(k)} - T\,{}^{B}\boldsymbol{p}_i \right\|^2 $$

4. **Check convergence.** Compute the registration error:
$$ \mathcal{E}^{(k)} = \frac{1}{N_p} \sum_{i=1}^{N_p} \left\| {}^{A}\boldsymbol{m}_i^{(k)} - {}^{A}T_{B}^{(k+1)}\,{}^{B}\boldsymbol{p}_i \right\|^2 $$

   If $\mathcal{E}^{(k)} < \varepsilon$ (a prescribed tolerance), stop; otherwise set $k \leftarrow k+1$ and return to step 1.

Properties of ICP.

- **Convergence guarantee**: ICP is guaranteed to converge monotonically (the error $\mathcal{E}^{(k)}$ is non-increasing at each iteration), but only to a *local* minimum of the objective function.
- **Sensitivity to initialisation**: a poor initial estimate can cause ICP to converge to a wrong local minimum (a misaligned configuration that is nonetheless locally optimal). This is why the Acrobot procedure, for example, acquires 4 landmark points first to initialise the algorithm before collecting the full surface point set.
- **Variants**: many extensions of ICP exist that improve robustness (e.g., point-to-plane ICP, trimmed ICP for partial overlaps, probabilistic variants). The basic version described here is the original Besl–McKay algorithm (1992).

6. Clinical Examples

The following six systems illustrate how the general registration framework specialises across different surgical domains. In each case, the registration chain linking the pre-operative plan to the robot frame is traced explicitly.

6.1. Example 1: Robodoc (Hip Replacement Surgery)

Robodoc is an autonomous milling robot for cementless hip replacement, developed in the late 1980s. Its registration procedure is a pure **3D/3D point-to-point** registration:

1. **Pre-operatively**: a set of n titanium fiducial screws is implanted into the femur of the patient under anaesthesia, several days before surgery. CT imaging is then performed; the 3D coordinates of the fiducial centroids are measured in the CT frame ($\mathcal{R}_{\text{patient}}$ in the pre-operative sense).
2. **Intra-operatively**: the bone is exposed. The robot's end-effector (equipped with a probing tip) physically touches each fiducial in turn, recording its 3D position in the robot frame $\mathcal{R}_{\text{robot}}$ using the robot's forward kinematics.
3. The known correspondence between each pre-operative fiducial coordinate and its intra-operative robot-frame coordinate provides the n point pairs needed to solve the Procrustes problem and compute ${}^{\text{patient}}T_{\text{robot}}$.

The fiducial screws guarantee high-precision, unambiguous correspondences. The main drawback is the invasiveness: a separate surgical procedure under anaesthesia is required before the main operation. Robodoc was the first clinical robot to use this registration paradigm, and the approach remains highly accurate.

6.2. Example 2: CASPAR (Knee Replacement Surgery)

CASPAR (Computer-Assisted Surgical Planning and Robotics) is similar in concept to Robodoc but adds intra-operative **motion detection** to handle patient movement during surgery:

- **Registration**: point-to-point using a set of implanted fiducials S, palpated by the robot to establish ${}^{\text{patient}}T_{\text{robot}}$.
- **Motion tracking**: a separate set of fiducials S' is mounted on a rigid clamp attached to the bone. An optical localizer continuously tracks S' throughout the procedure. If the patient moves (e.g., due to muscle relaxation or surgeon manipulation), the tracker detects the displacement and the robot controller compensates accordingly, maintaining registration accuracy without requiring re-registration.

The distinction between S (for registration) and S' (for motion tracking) is important: registration establishes the initial frame relationship, while tracking maintains it in real time.

6.3. Example 3: Acrobot (Knee Replacement Surgery)

Unlike Robodoc and CASPAR, the Acrobot uses a **surface-based** (non-invasive) registration strategy, avoiding the implantation of fiducial screws:

1. **Pre-operatively**: a 3D surface model of the knee bones is reconstructed from CT data. The surgical plan (the flat bone cuts) is expressed in this model's frame.
2. **Intra-operatively, initialisation**: the surgeon uses a tracked pointer to palpate **4 anatomical landmark points** on the exposed bone surface. These landmarks are also identifiable in the CT model, providing an initial point-to-point registration to initialise the ICP algorithm.
3. **ICP registration**: the surgeon then palpates a set of **20–30 randomly selected points** on the bone surface. The ICP algorithm matches this intra-operative point cloud to the pre-operative CT surface model, refining the registration.
4. **Validation**: the registration accuracy is verified by displaying the palpated points overlaid on the bone surface model. If the accuracy is insufficient, the procedure is repeated with a new set of points.
5. **Motion compensation**: femoral and tibial clamps form a rigid body with the bone and provide known hole positions, enabling rapid re-registration if patient motion occurs.

The accuracy of the Acrobot's registration procedure was formally evaluated and reported in the IEEE-TRA 2003 paper by Jakopec et al.

> **Trade-off: fiducials vs. surface registration.** Fiducial-based registration (Robodoc, CASPAR) is more accurate and less sensitive to initialisation, but requires an invasive pre-operative implantation procedure. Surface-based registration (Acrobot) is non-invasive but requires a good initialisation (the 4 landmark points) and can be less accurate if the surface point set is poorly distributed or noisy. The choice depends on the clinical context: the higher accuracy demands of femoral stem milling in hip replacement historically justified the invasiveness of fiducial screws.

6.4. Example 4: Physical–Digital Twins (Rome Technopole PNRR Project)

This example is a current research project at Sapienza, funded by Rome Technopole under the Italian PNRR programme, focused on **surgical training and planning** using physical–digital twin technology.

The system consists of three layers:

- **Physical layer**: an anatomical phantom (a physical model of a cervical vertebra and surrounding soft tissue). The operator interacts with the phantom using a tracked stylus and wears a Meta Quest 3 mixed-reality headset.
- **Digital layer**: a Unity-based software environment maintains a real-time digital twin of the phantom, including deformable muscle geometry. The digital model is rendered on the screen and in the headset.
- **Registration**: the reference frame of the digital model must be aligned with the reference frame of the physical phantom as seen by the optical localizer. This is achieved by a **point-to-point registration**: the operator touches a set of known landmark points on the physical phantom with the stylus; the localizer records their coordinates in its frame; these are matched to the corresponding coordinates in the digital model's frame to compute the registration transform.

Once registered, the system allows the operator to touch the physical phantom with the stylus and see, in the headset, the corresponding position in the digital anatomy — including internal structures (spinal cord, vertebral arteries, nerve branches) that are not visible on the physical surface. This has direct training value: the operator can practise a surgical gesture (e.g., pedicle screw insertion) while receiving real-time feedback about proximity to critical structures.

The system also demonstrates **tool tracking** in real time: the pose of the stylus is continuously updated by the localizer and reflected in the digital scene. The practical session accompanying this lecture gave students the opportunity to perform the registration procedure themselves and interact with the registered system.

> **Deformation matching.** One limitation highlighted in the lecture is that full volumetric deformation simulation is computationally expensive and not directly supported by the Unity physics engine. Currently, only surface (cloth) deformation is modelled, not volumetric deformation of muscles and connective tissue. Extending the digital twin to support volumetric soft-tissue simulation is an active research challenge.

6.5. Example 5: Speedy (Neurosurgery)

Speedy is a neurosurgical robot (developed in Grenoble, J. Troccaz group) for stereotactic targeting. Its registration chain is more complex, involving multiple imaging modalities and calibration steps:

- **Pre-operatively**: the surgical plan is defined on CT images in the patient planning frame $\mathcal{R}_{\text{patient}}$. Pre-operative images from different modalities (CT and MRI) are registered together using anatomical landmarks for the pre-op/intra-op step and marker-based registration for intra-op/intra-op alignment.
- **X-ray calibration**: the intrinsic geometric model of the intra-operative X-ray system is calibrated using a calibration cage with known geometry, relating the 2D image frame to 3D space.
- **X-ray/robot calibration**: the transformation between the X-ray imaging frame and the robot base frame is established via a direct calibration procedure (placing known markers visible in both the X-ray image and the robot's kinematic model).
- **Intra-operative registration**: at the start of surgery, an intra-operative X-ray image is acquired. Fiducial markers visible in this image are matched to their pre-operative CT coordinates (marker-based registration), establishing ${}^{\text{patient/planning}}T_{\text{patient/OR}}$.

The complete chain of transformations — pre-operative image frame → patient/OR frame → X-ray frame → robot frame — then allows the pre-operative plan to be expressed in the robot frame.

6.6. Example 6: CyberKnife and CyberKnife+Synchrony (Radiosurgery)

CyberKnife is a robotic radiosurgery system that mounts a linear accelerator (LINAC) on a serial robot arm and delivers precisely aimed radiation beams to tumours anywhere in the body, without physical incision.

6.6.1. CyberKnife – Isocentric Mode

In the basic isocentric mode, all beams are directed through a common isocentre. The registration chain involves:

- **Pre-operatively**: radiation dose planning on CT images, defining the beam directions and intensities in the patient planning frame.
- **X-ray calibration**: two orthogonal X-ray sources (imaging sensors 1 and 2) are calibrated with respect to the isocentre frame $\mathcal{R}_{\text{isocenter}}$.
- **Robot/isocentre calibration**: the robot base frame $\mathcal{R}_{\text{robot}}$ is related to the isocentre frame.
- **Intra-operative image registration**: iterated registration of intra-operative X-ray images to the pre-operative CT (DRR — Digitally Reconstructed Radiograph — matching) establishes ${}^{\text{patient/planning}}T_{\text{patient/OR}}$.
- **Set-up**: the patient is positioned on the treatment couch; the couch frame $\mathcal{R}_{\text{isocenter/table}}$ provides the final link.

6.6.2. CyberKnife+Synchrony – Non-Isocentric Mode with Respiratory Tracking

The Synchrony extension addresses targets (e.g., lung tumours) that move with respiration. It combines two types of fiducials:

- **Internal fiducials** (gold seeds): radio-opaque gold markers surgically implanted near the tumour. They are visible in intra-operative X-ray images and provide the initial registration between the pre-operative CT and the intra-operative X-ray frame.

- **External fiducials** (infrared diodes): optical markers placed on the patient's skin that move with respiration. They are tracked by the localizer in real time.

A **learned model** relating internal tumour position (from periodic X-ray snapshots) to external marker position (from continuous optical tracking) allows the system to predict the tumour's intra-breath position at all times without continuous X-ray irradiation. The robot continuously adjusts the LINAC orientation to follow the tumour, making the treatment robust to respiratory motion.

7. The Point-to-Point Registration Problem: Closed-Form Solution

For completeness, the closed-form solution to the point-to-point (Procrustes) problem with known correspondences is summarised here. Given n corresponding point pairs $\{({}^A\boldsymbol{f}_i,\ {}^B\boldsymbol{f}_i)\}_{i=1}^n$, the optimal rigid transformation is found as follows:

1. **Compute centroids.**

$$\bar{\boldsymbol{f}}^A = \frac{1}{n}\sum_{i=1}^{n} {}^A\boldsymbol{f}_i, \qquad \bar{\boldsymbol{f}}^B = \frac{1}{n}\sum_{i=1}^{n} {}^B\boldsymbol{f}_i$$

2. **Centre the point sets.**

$$\tilde{\boldsymbol{f}}_i^A = {}^A\boldsymbol{f}_i - \bar{\boldsymbol{f}}^A, \qquad \tilde{\boldsymbol{f}}_i^B = {}^B\boldsymbol{f}_i - \bar{\boldsymbol{f}}^B$$

3. **Compute the cross-covariance matrix.**

$$H = \sum_{i=1}^{n} \tilde{\boldsymbol{f}}_i^B \left(\tilde{\boldsymbol{f}}_i^A\right)^T \in \mathbb{R}^{3\times 3}$$

4. **SVD decomposition.** Compute $H = U\Sigma V^T$.
5. **Optimal rotation and translation.**

$${}^A R_B^* = VU^T, \qquad {}^A\boldsymbol{t}_B^* = \bar{\boldsymbol{f}}^A - {}^A R_B^*\, \bar{\boldsymbol{f}}^B$$

   If $\det(VU^T) = -1$ (reflection rather than rotation), replace the last column of V with its negation to enforce ${}^A R_B^* \in SO(3)$.

This solution is exact (not iterative) and globally optimal for the least-squares criterion. It forms the inner step of the ICP algorithm (Step 3 in Section 4.2).

8. Summary and Key Conceptual Threads

- **Registration, calibration, and tracking are distinct operations.** Calibration relates fixed devices once; tracking updates a moving tool's pose continuously; registration relates the pre-operative plan to the intra-operative robot frame, once per procedure (with possible re-registration).
- **All registrations require features.** Features can be artificial (implanted fiducials, attached markers) or natural (anatomical landmarks, palpated surface points). The choice involves a trade-off between invasiveness and accuracy.
- **Known vs. unknown correspondences.** Point-to-point registration assumes known correspondences and yields a closed-form SVD solution. Point-to-surface registration uses ICP to iteratively estimate correspondences and is guaranteed to converge to a local minimum — making a good initialisation essential.

- **The complexity of the registration chain scales with the complexity of the procedure.** Orthopaedic systems (Robodoc, Acrobot) require a single pre-op/intra-op registration. Neurosurgical systems (Speedy) require multiple calibrations and registrations across imaging modalities. Radiosurgical systems (CyberKnife+Synchrony) additionally require real-time registration to compensate for respiratory motion.
- **Patient motion invalidates registration.** Any movement of the patient after registration requires re-registration or continuous tracking-based compensation (as in CASPAR and CyberKnife+Synchrony). This is a fundamental operational constraint in all robot-assisted surgery.

Key topics for revision.

1. The definitions of registration, robot registration, calibration, and tracking, and the differences among them.
2. The mathematical formulation of rigid 3D/3D registration: features, distance measure, optimisation problem.
3. Point-to-point registration (Procrustes problem): minimum number of points, SVD-based closed-form solution, use of fiducials vs. anatomical landmarks.
4. Point-to-surface registration and ICP: the four-step algorithm, the convergence guarantee (local minimum), sensitivity to initialisation.
5. The registration procedures of Robodoc (implanted fiducials, direct palpation), CASPAR (fiducials + motion tracking), and Acrobot (surface ICP with landmark initialisation).
6. The physical–digital twin system: layers, registration method, role of optical localizer, limitation of volumetric deformation.
7. Speedy: multi-frame registration chain across imaging modalities.
8. CyberKnife+Synchrony: internal/external fiducials, learned motion model, real-time respiratory tracking.

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